CS 306: Digital Transmission

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Outline

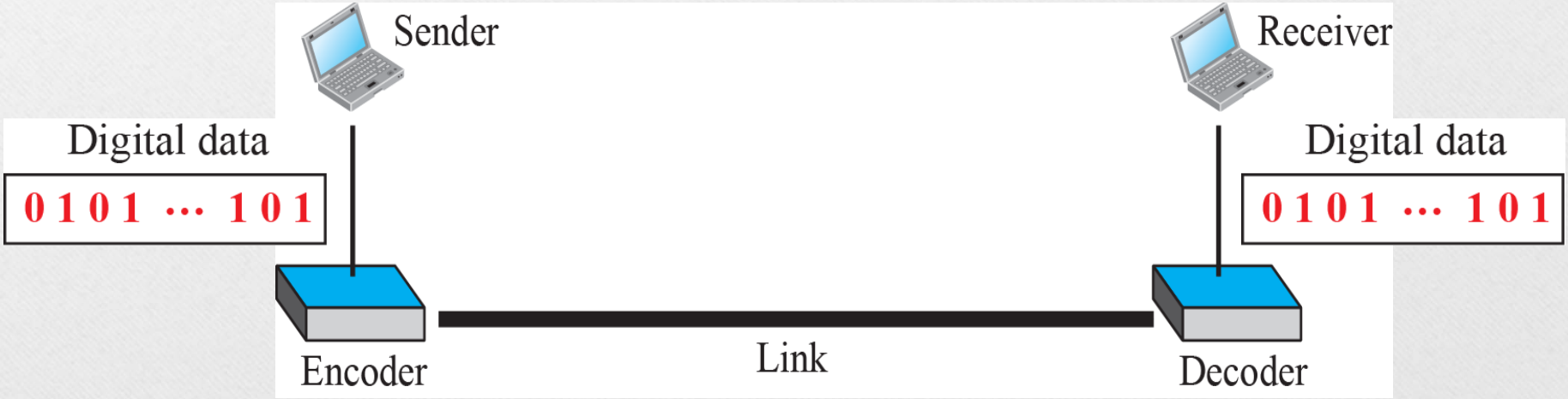
- Digital-to-Digital Conversion
- Analog-to-Digital Conversion
- Transmission Modes

Digital-to-Digital Conversion

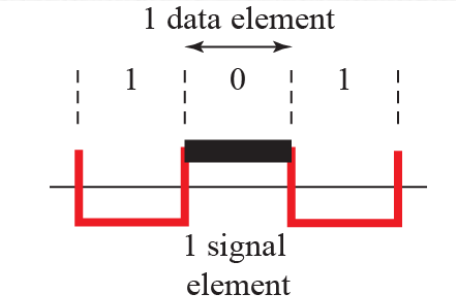
- The conversion of digital data to digital signal involves three techniques:
  - Line coding
  - Block coding
  - Scrambling.
- Line coding is always needed; block coding and scrambling may or may not be needed.

Line Coding

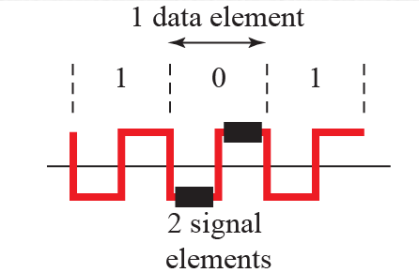
- Line coding converts a sequence of bits to a digital signal. At the sender, digital data are encoded into a digital signal; at the receiver, the digital data are recreated by decoding the digital signal.



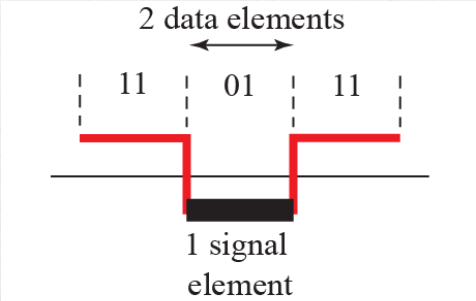
Signal and Data Elements



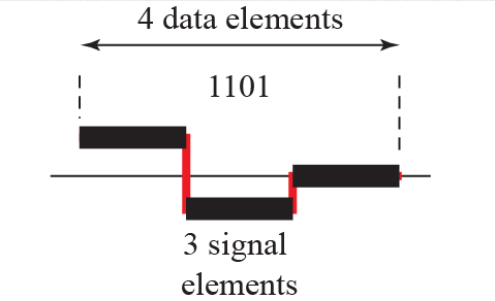
a. One data element per one signal element (r = 1)



b. One data element per two signal elements $\left(r = \frac{1}{2}\right)$

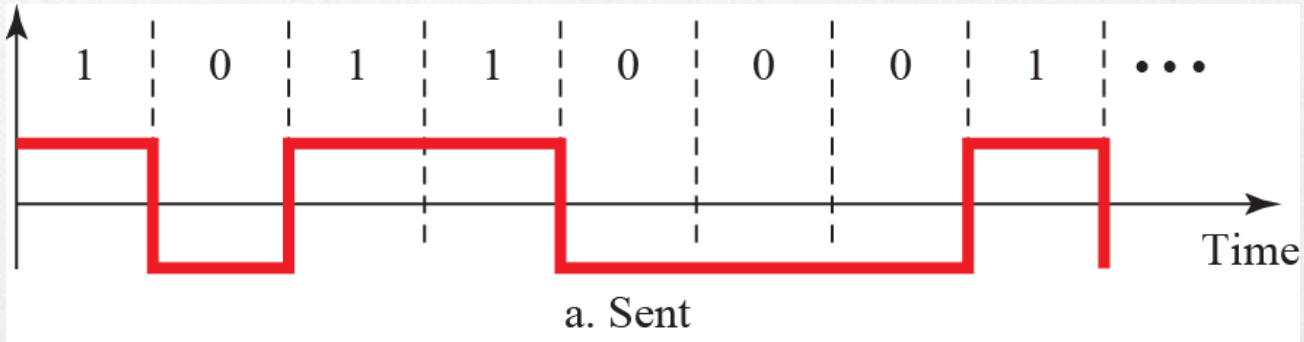


c. Two data elements per one signal element (r = 2)

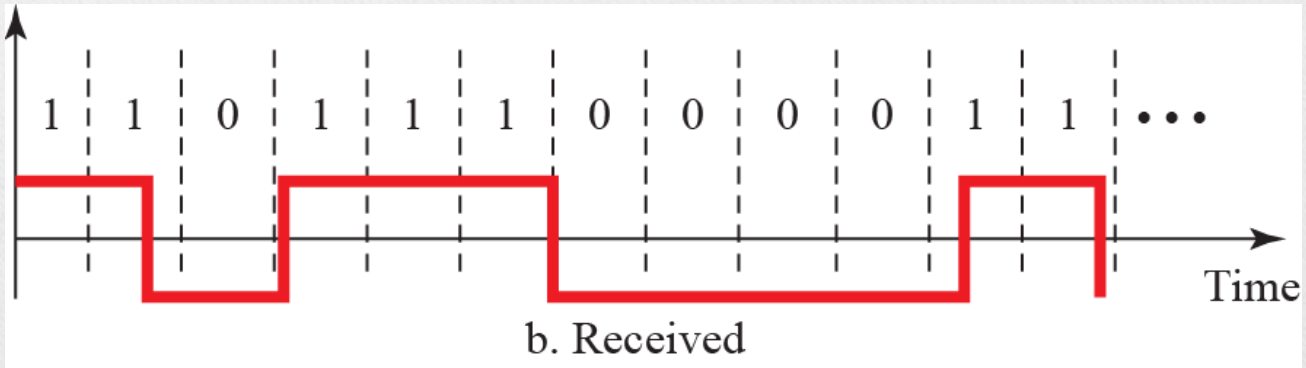


d. Four data elements per three signal elements $\left(r = \frac{4}{3}\right)$

Effect of lack of synchronization

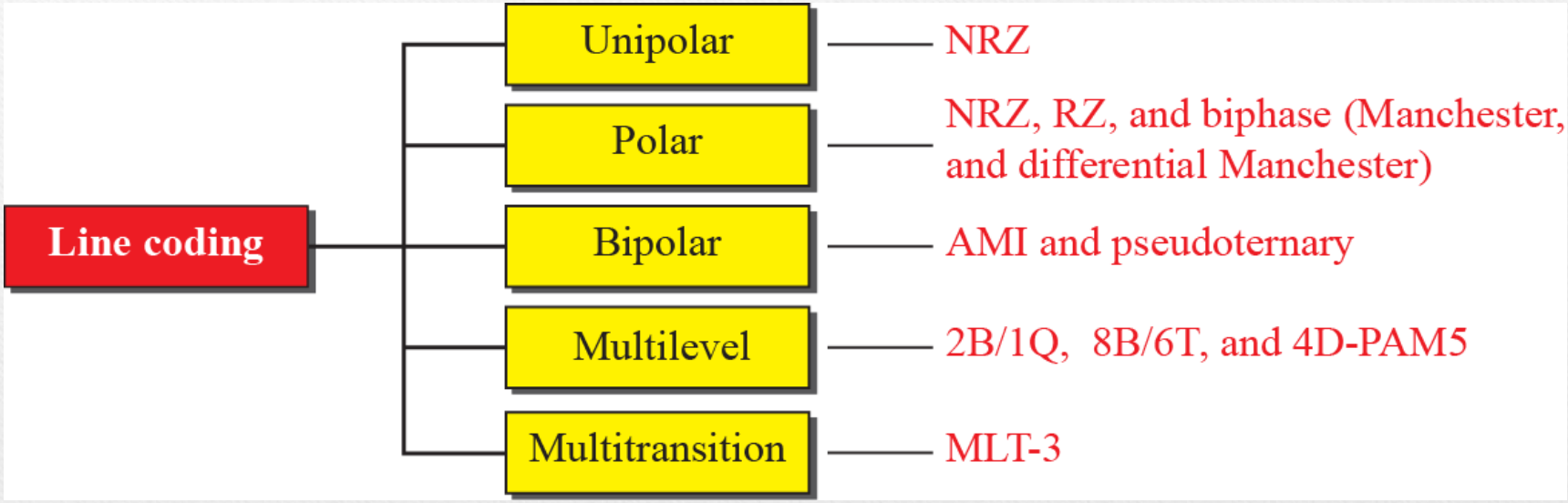


a. Sent

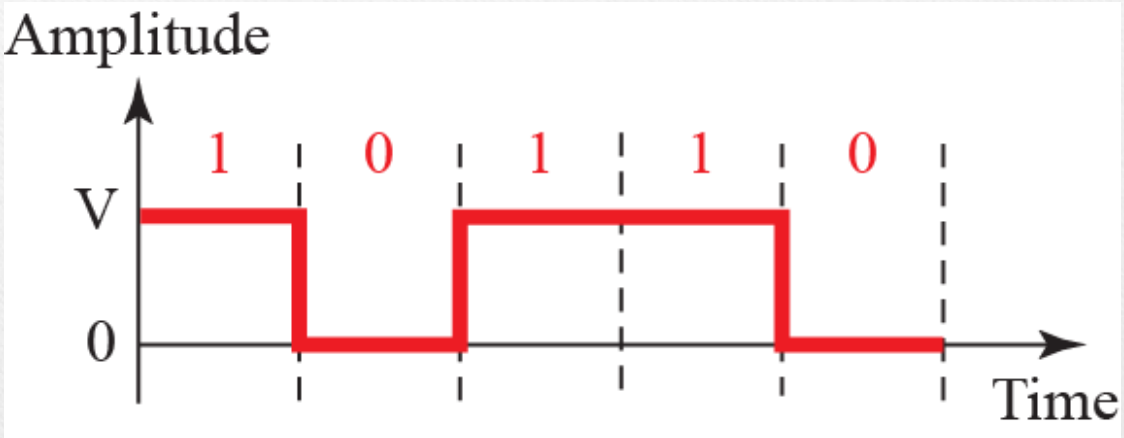


b. Received

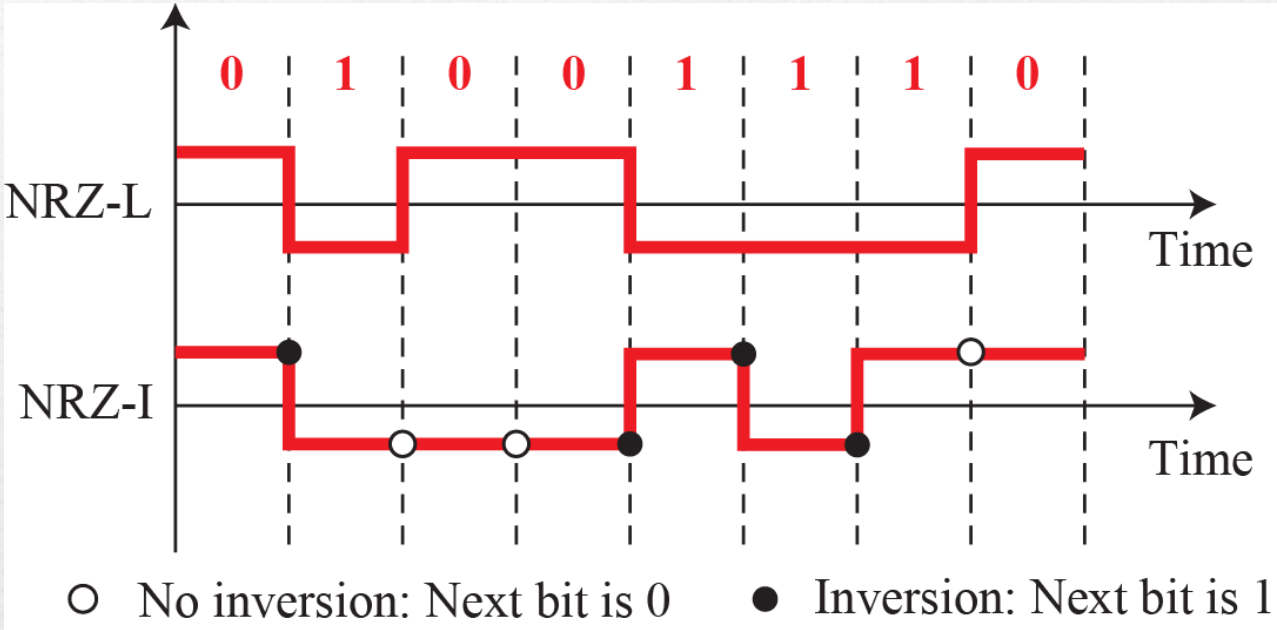
Line Coding Schemes



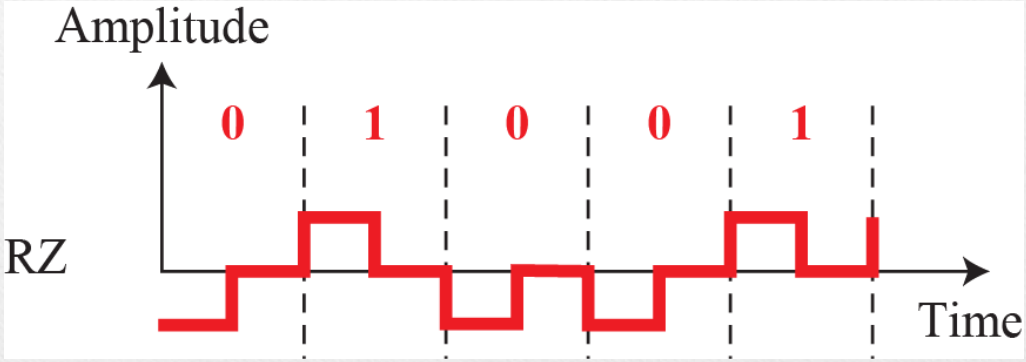
Unipolar NRZ scheme



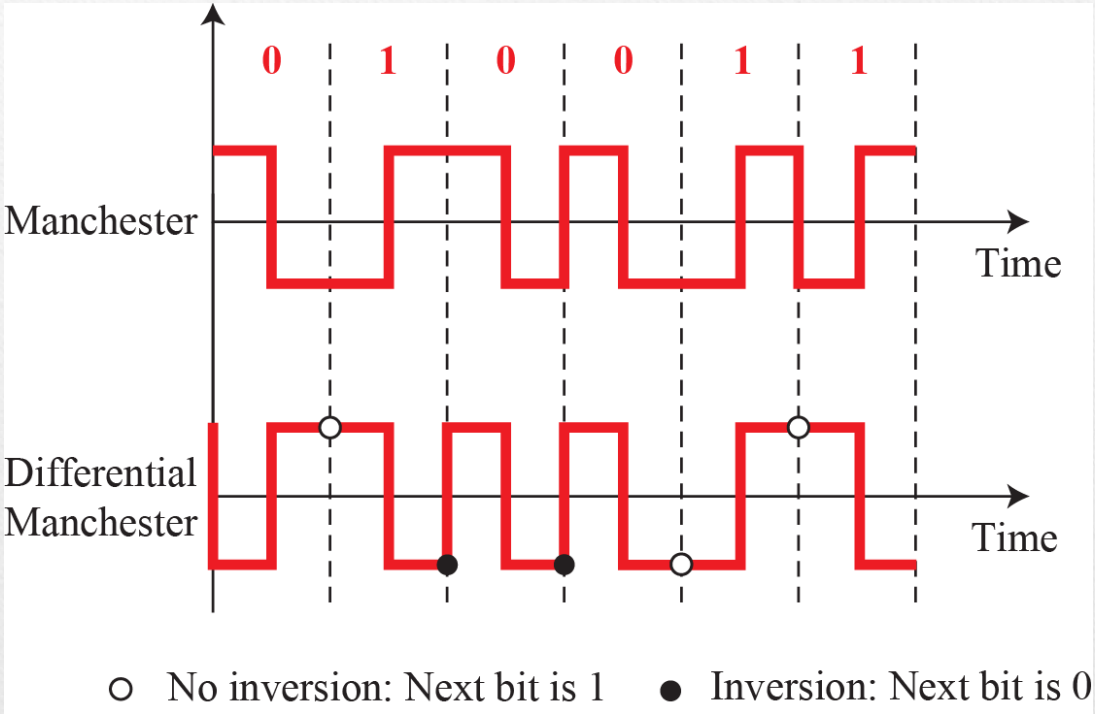
Polar schemes (NRZ-L and NRZ-I)



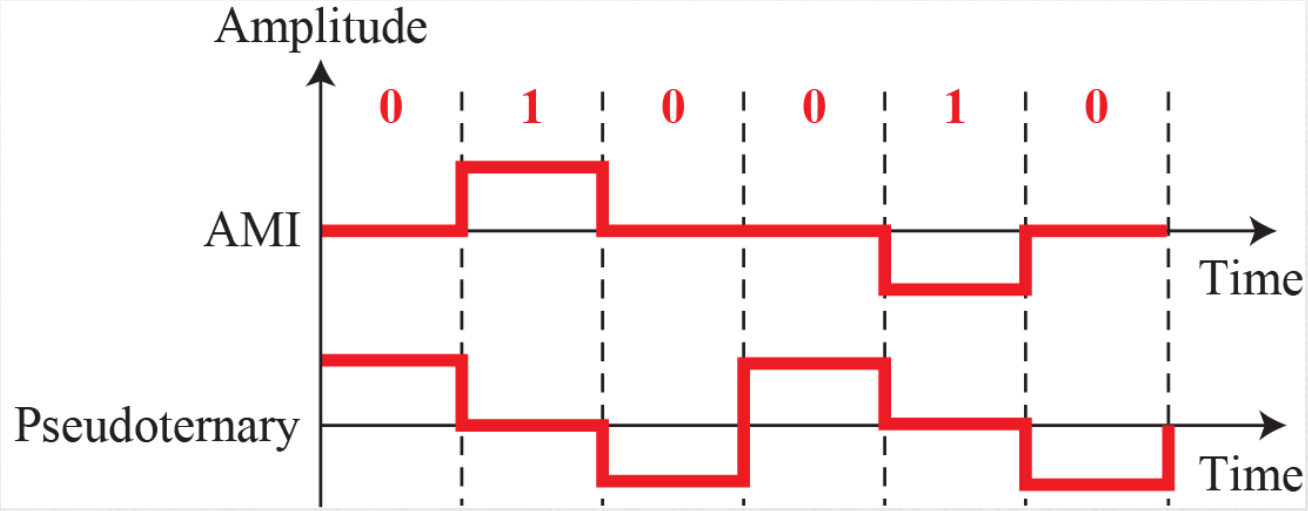
Polar schemes (RZ)



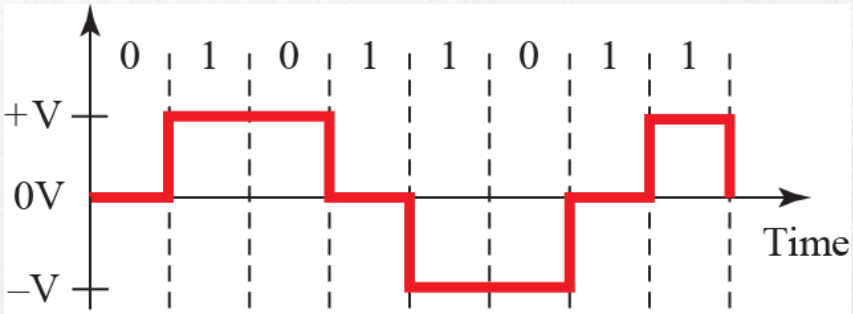
Polar Biphase



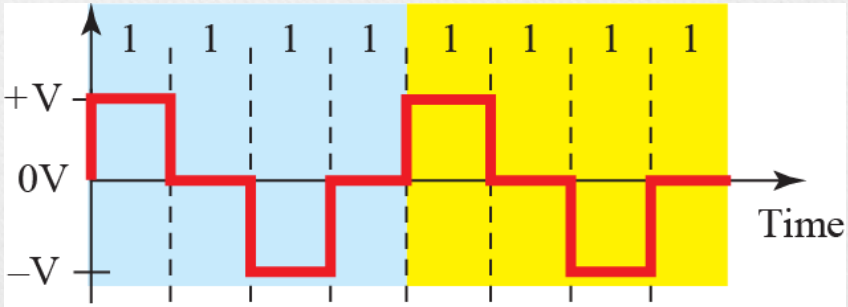
Bipolar schemes



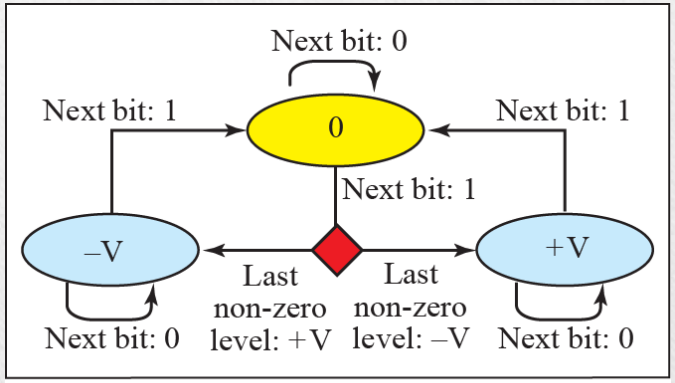
Multi-Transition MLT-3 Scheme



a. Typical case



b. Worst case



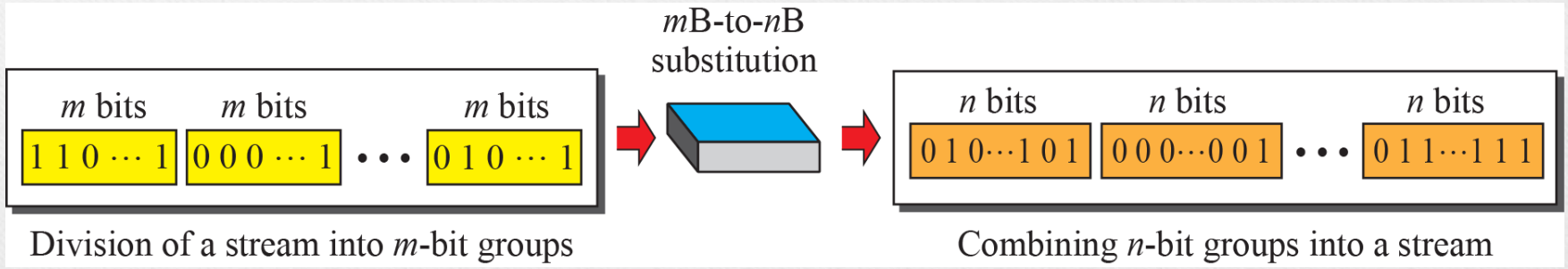
c. Transition states

Summary of line coding schemes

Category	*Scheme*	*Bandwidth (average)*	*Characteristics*
Unipolar	NRZ	$B = N/2$	Costly, no self-synchronization if long 0s or 1s, DC
Polar	NRZ-L	$B = N/2$	No self-synchronization if long 0s or 1s, DC
	NRZ-I	$B = N/2$	No self-synchronization for long 0s, DC
	Biphase	$B = N$	Self-synchronization, no DC, high bandwidth
Bipolar	AMI	$B = N/2$	No self-synchronization for long 0s, DC
Multilevel	2B1Q	$B = N/4$	No self-synchronization for long same double bits
	8B6T	$B = 3N/4$	Self-synchronization, no DC
	4D-PAM5	$B = N/8$	Self-synchronization, no DC
Multitransition	MLT-3	$B = N/3$	No self-synchronization for long 0s

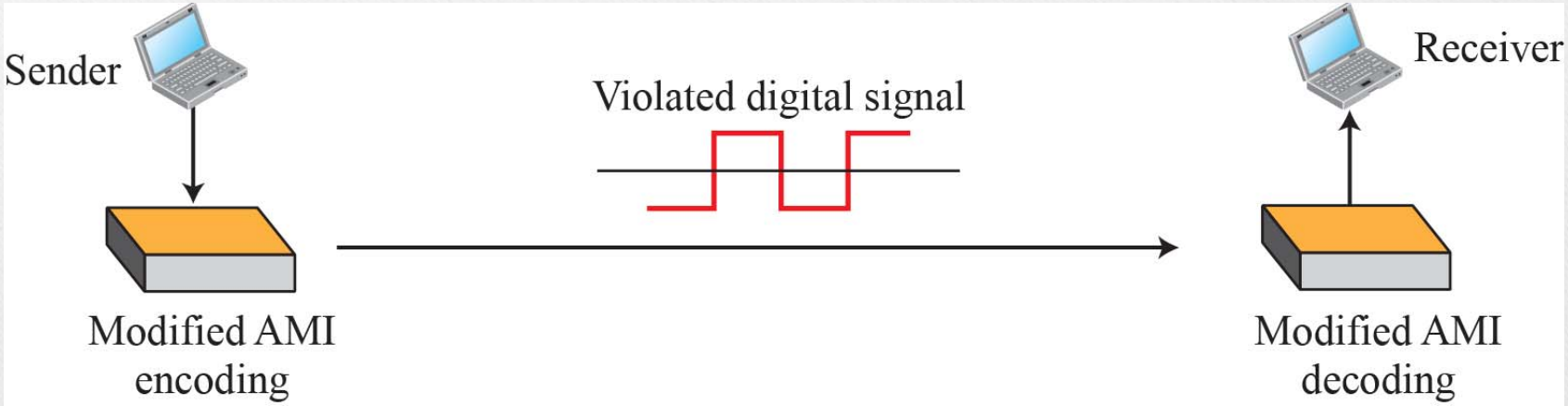
Block Coding

- Block coding changes a block of m bits into a block of n bits, where n is larger than m. Block coding is referred to as an mB/nB encoding technique



Scrambling

- The system needs to insert the required pulses based on the defined scrambling rules.

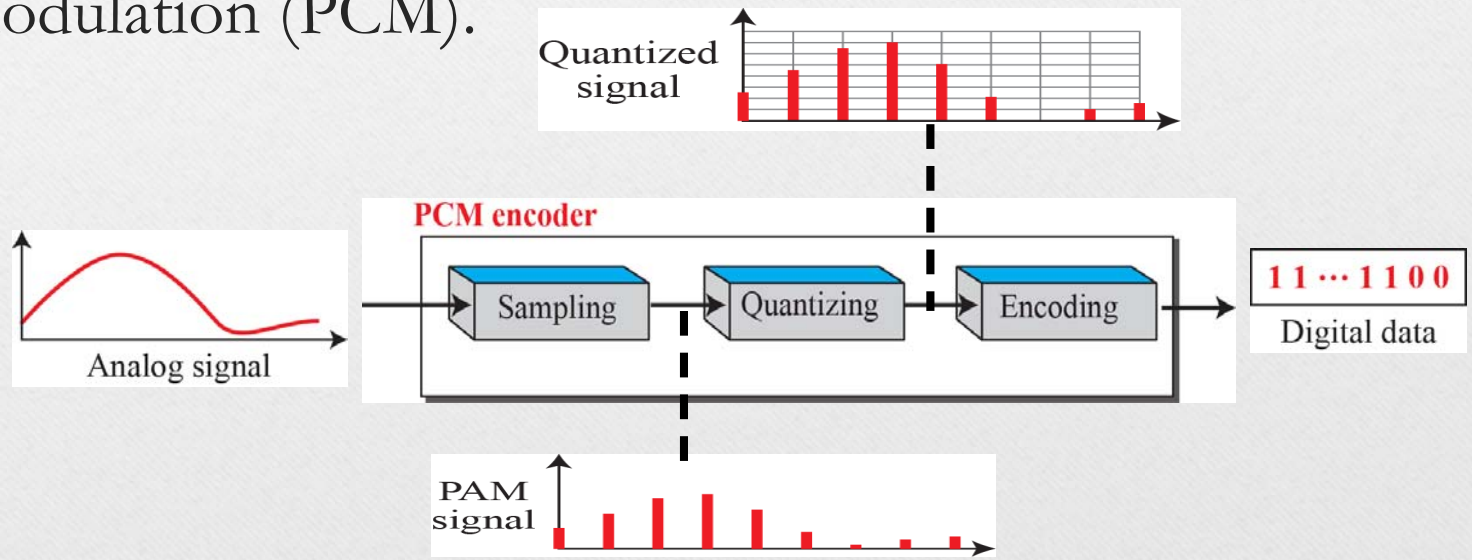


Analog-to-Digital Conversion

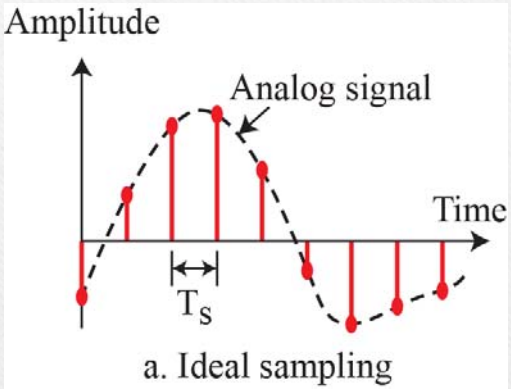
- To change an analog signal to digital data, we have two techniques:
  - Pulse code modulation
  - Delta modulation.

Pulse Code Modulation (PCM)

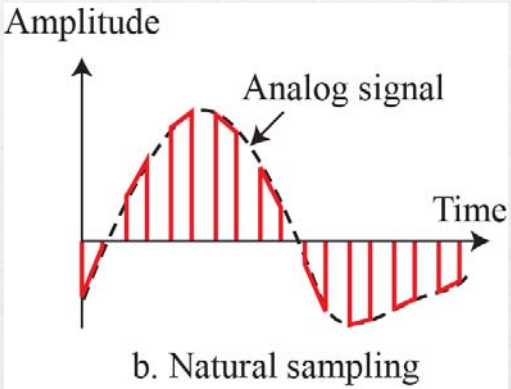
- The most common technique to change an analog signal to digital data (digitization) is called pulse code modulation (PCM).



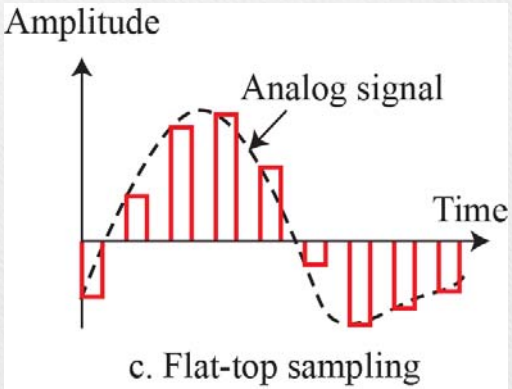
Three different sampling methods for PCM



a. Ideal sampling

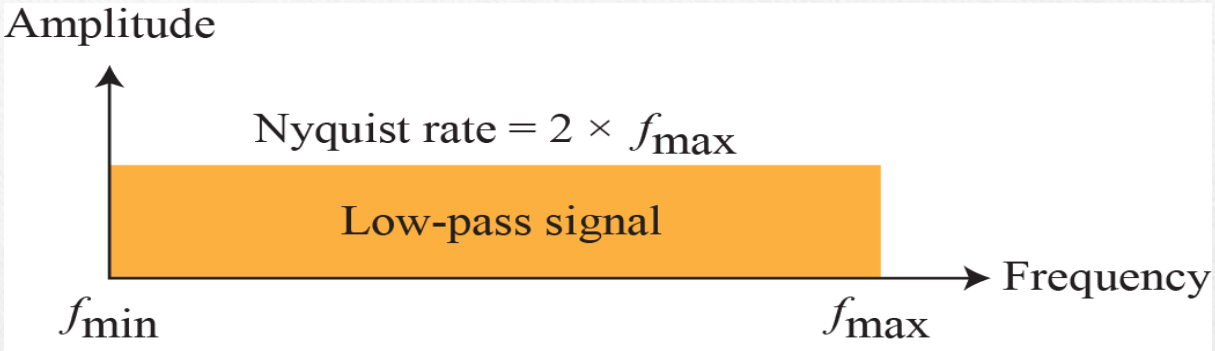


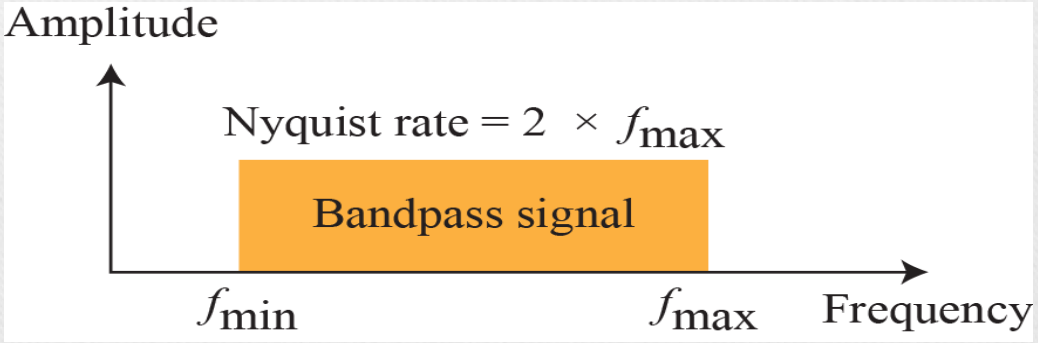
b. Natural sampling



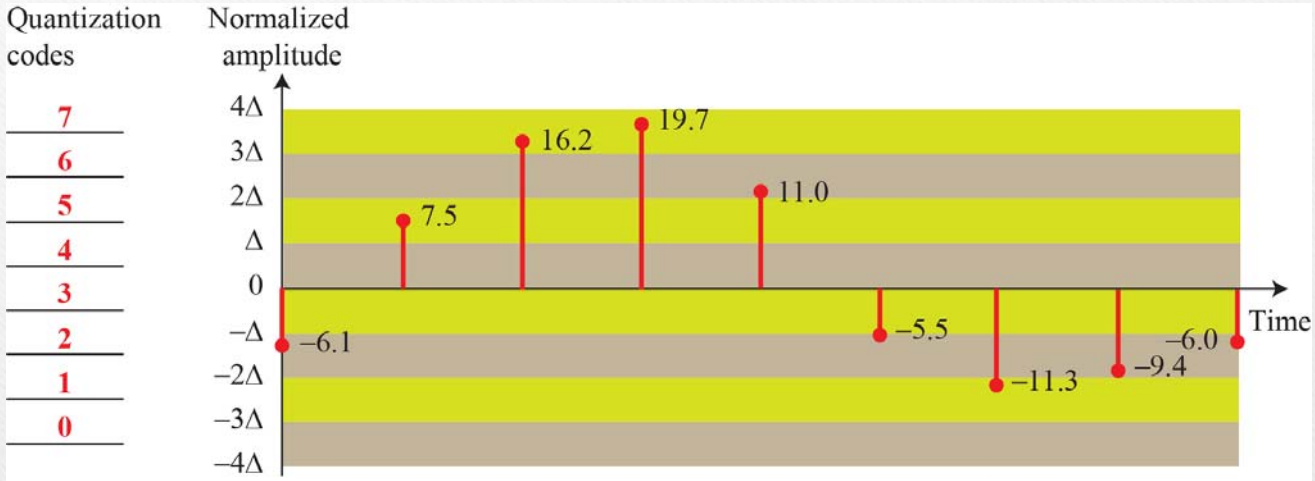
c. Flat-top sampling

Nyquist Sampling Rate for Lowpass and Bandpass Signals



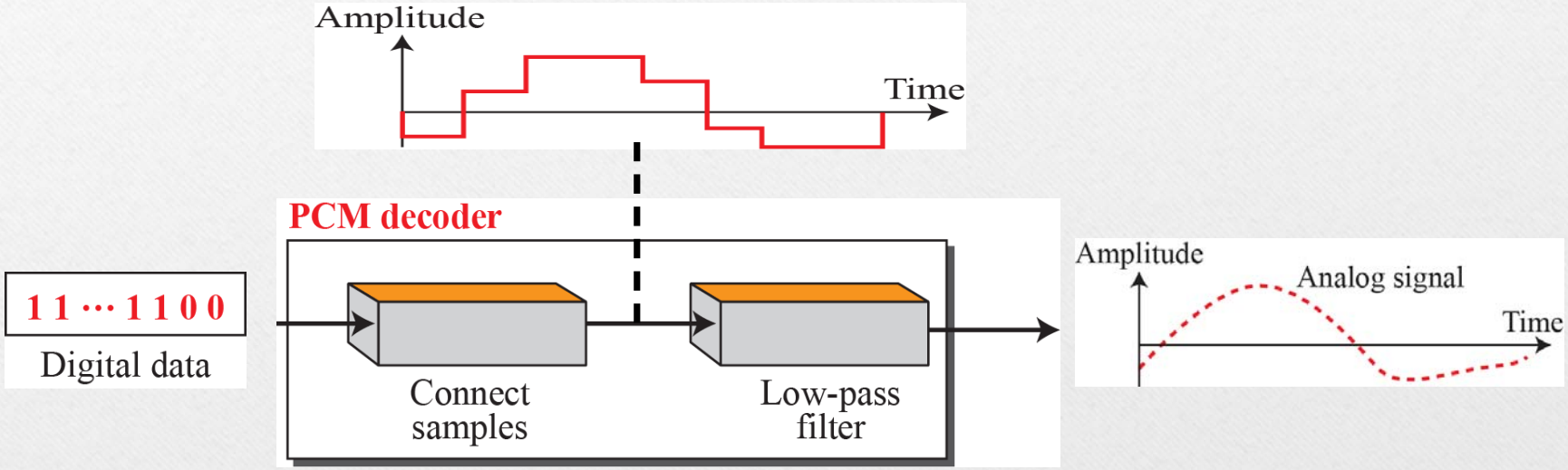


Quantization and Encoding of A Sampled Signal



Normalized PAM values	−1.22	1.50	3.24	3.94	2.20	−1.10	−2.26	−1.88	−1.20
Normalized quantized values	−1.50	1.50	3.50	3.50	2.50	−1.50	−2.50	−1.50	−1.50
Normalized error	−0.28	0	+0.26	−0.44	+0.30	−0.40	−0.24	+0.38	−0.30
Quantization code	2	5	7	7	6	2	1	2	2
Encoded words	010	101	111	111	110	010	001	010	010

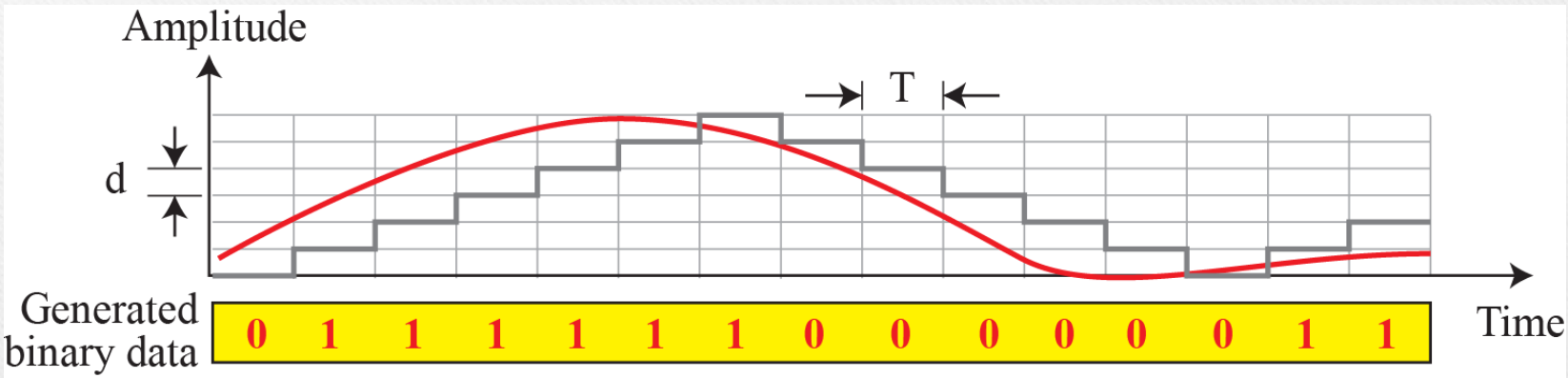
Components of a PCM Decoder



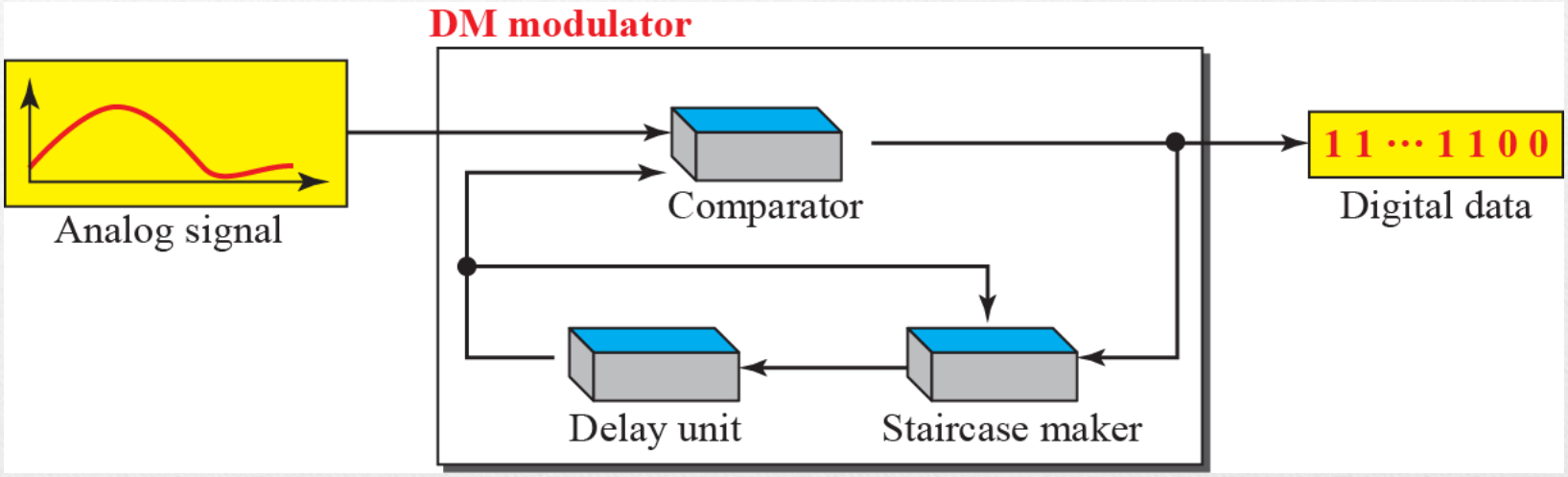
Delta Modulation (DM)

- PCM is a very complex technique. Other techniques have been developed to reduce the complexity of PCM.
- The simplest is delta modulation.
- PCM finds the value of the signal amplitude for each sample; DM finds the change from the previous sample.

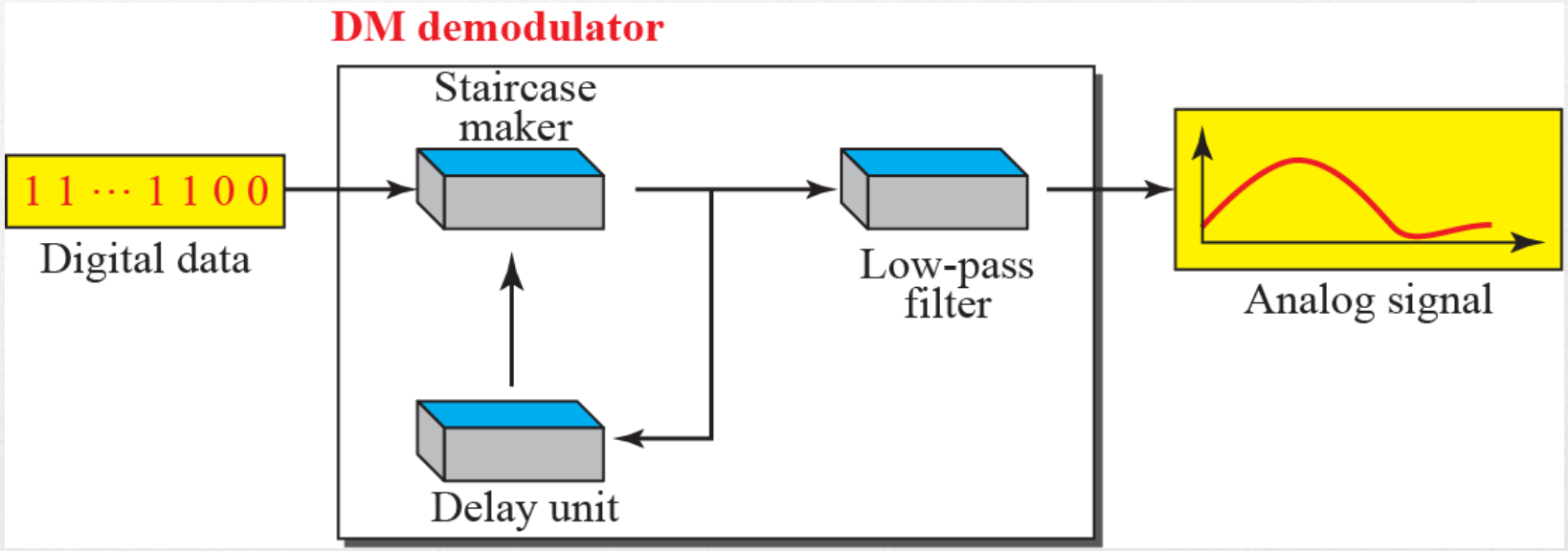
The Process of Delta Modulation



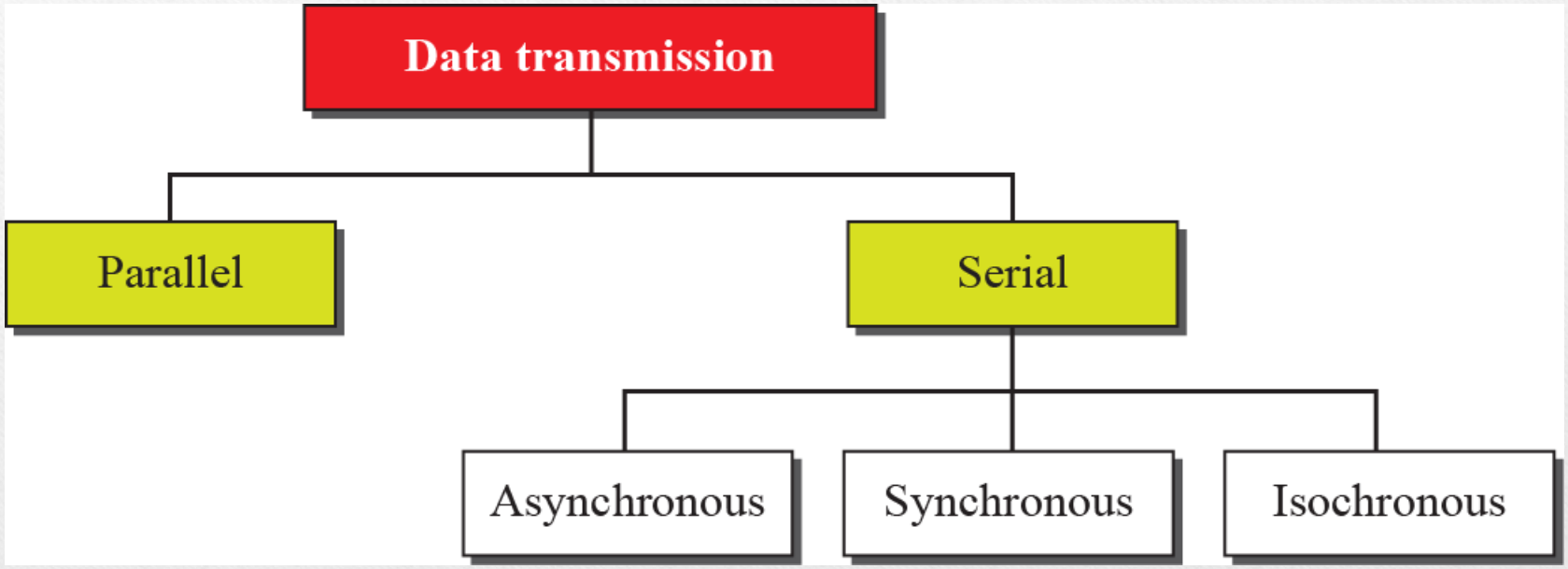
Delta Modulation Components



Delta Demodulation Components



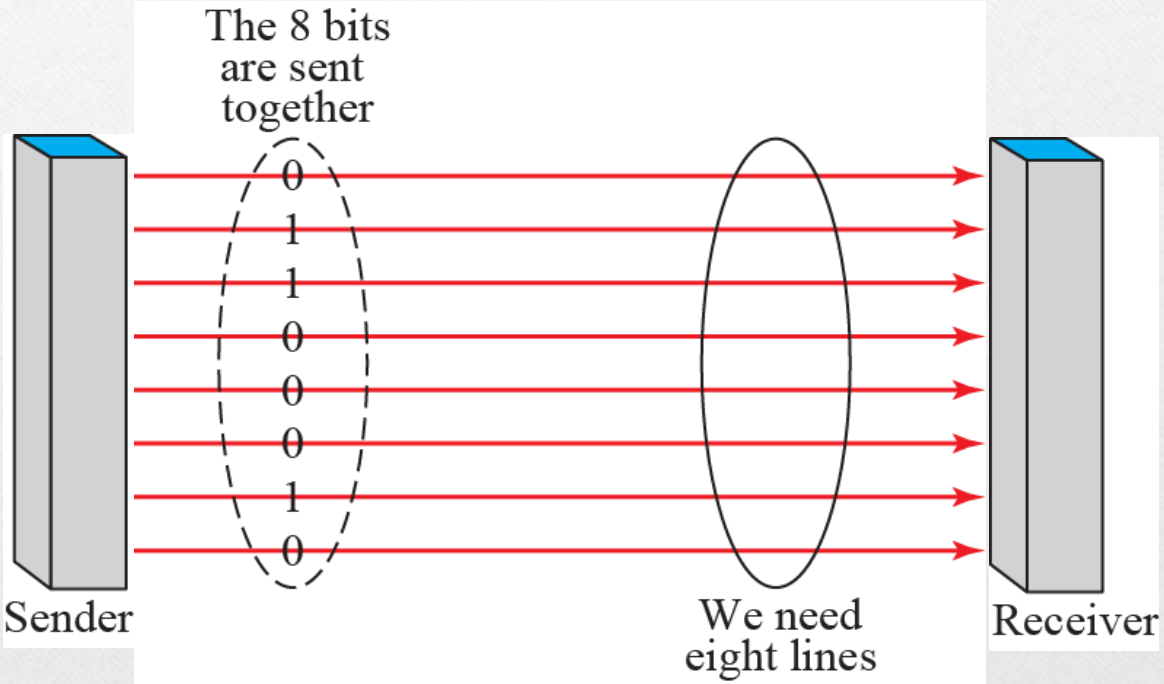
Data Transmission Modes



Parallel Transmission

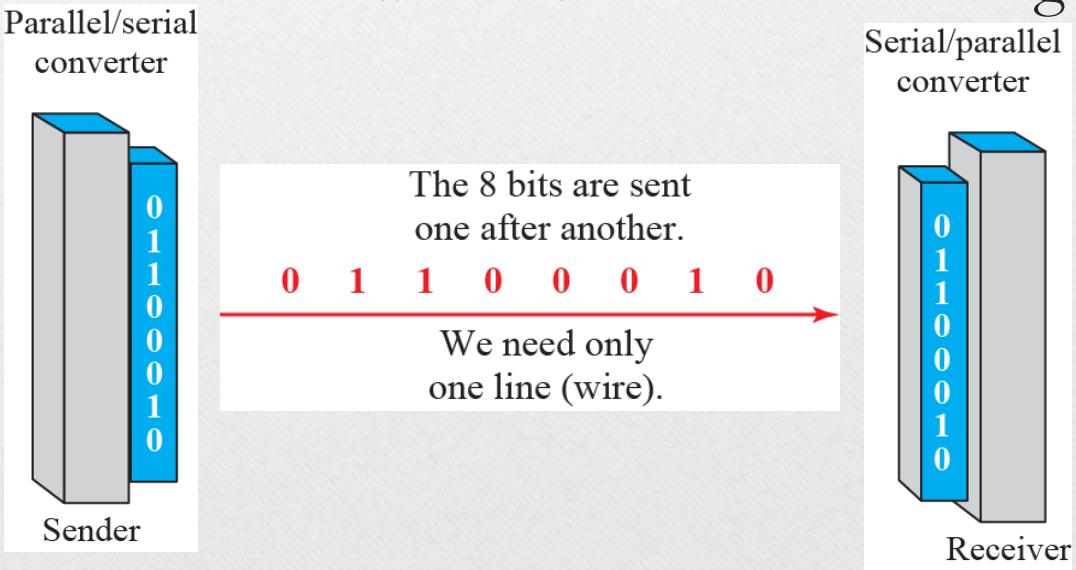
- Line coding converts a sequence of bits to a digital signal.
- At the sender, digital data are encoded into a digital signal; at the receiver, the digital data are recreated by decoding the digital signal.

Parallel Transmission (Contd.)

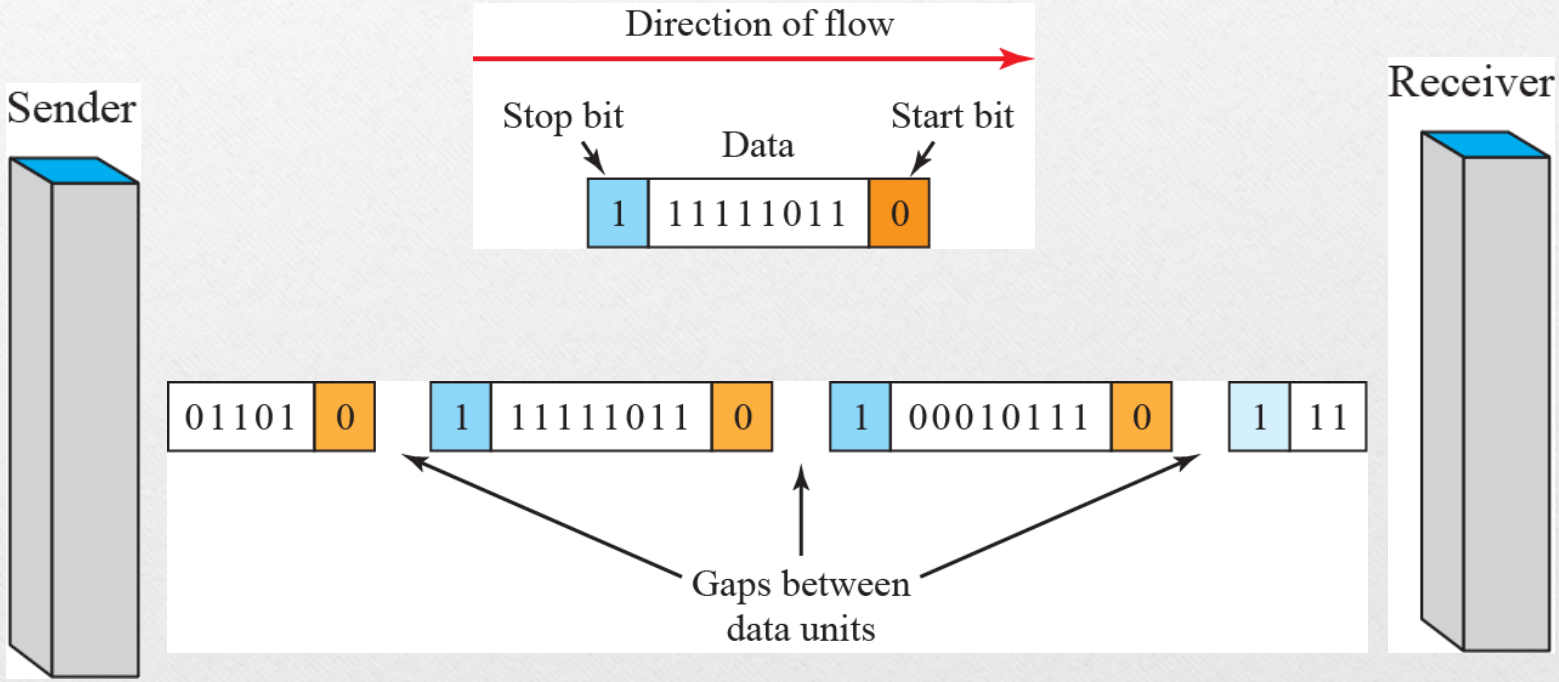


Serial Transmission

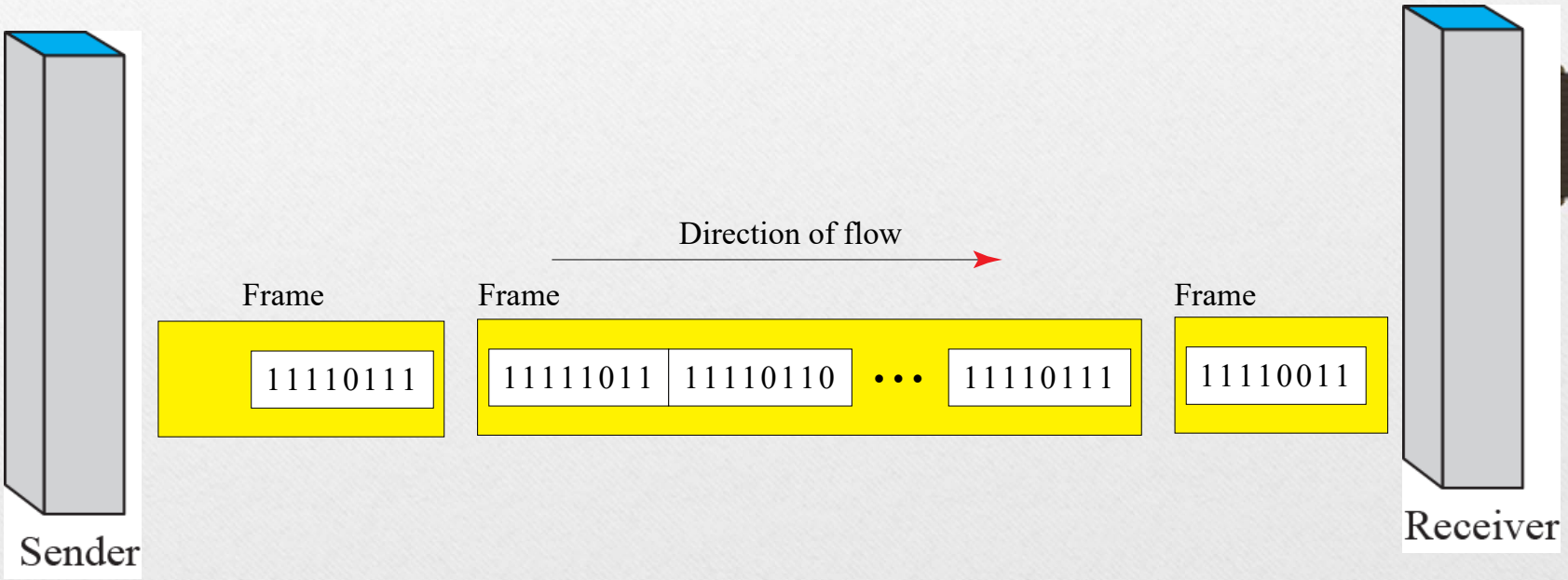
- In serial transmission one bit follows another, so we need only one communication channel rather than n to transmit data between two communicating devices



Asynchronous Transmission



Synchronous Transmission



Next Class

- Analog Transmission

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